JAMMING DETECTION IN PROVIDING FOR RADAR JAMMING IMMUNITY

Evgeny Markin, Member

Abstract: **The article considers the objective of providing for radar jamming immunity by optimizing the radar structure, parameters or operating algorithms on the basis of correct recognition of jamming in progress. Procedure of recognition in the determined feature space is examined. Viability of this approach is demonstrated using the example of airborne radar operation against the source of polarization and blinking jamming.**

Index Terms: **radar, jamming immunity indicator, cross-polarization jamming, blinking jamming, repeat jamming, antipode-type interference.**

I. INTRODUCTION

Development of air assault weapons and mass-scale deployment of measures and aids to suppress the enemy air defence in air combat operations highlights the demand for creating jammer resistant radar systems. Viewed from this standpoint, the weakest among radar systems with semi-active and active homing channels appears to be the airborne radar of antiaircraft guided missile seeker. Under conditions of jamming impacts, a vital issue for the radar is to provide for stable automatic target tracking by angular coordinates. It is reasonable to characterize the quality of automatic tracking by the following criteria:

$$Q = \int_{0}^{T} e^2(t)dt \tag{1}$$

where [0,T] analysis interval, $e^2(t)$ - squared error of angle tracking. The anti-jamming capability issue boils down to the necessity to provide for best in the sense of precision (1) - target angle tracking under active jamming impacts which adversely affect radar accuracy characteristics. Two decision-making strategies seem to be relevant here: tradeoff and active ones. The first one can be characterized by the fact that prior to jamming impacts specified by the jammer class code $\{H\}$, the radar itself should possess certain qualities which unequivocally provide for reduction in efficiency of jammers on their deployment. For example, when the level of receiving antenna side lobes is less than $-48dB$, the covering jamming will be ineffective. Reduction of side lobes level for airborne radar with missile cross dimensions specified, however, would inevitably result in expansion of radar main beam and, consequently, in its worse resolution of angle coordinates. This would slow down acquisition of reference-input signals and reduce accuracy of target automatic angle tracking. To operate under conditions of blinking in proghress, the radar, in contrast, would require high angle resolution [1]. As a result, characteristics of airborne radar should be selected with the tradeoff in mind, making allowance, from the start, for some drop in Q quality at target tracking under conditions of each jamming impact. It is suggested here that the active strategy should be regarded as adaptive procedure (set of adaptive procedures) which make it possible to maintain Q quality criterion at the level not less than the specified one, when affected by q number of jammer classes H_i, $\forall i = \overline{1, q}$. In these conditions, adaptation can be of any shape: parametric, algorythmic, structural etc. It is important here that under impact of jamming signal H_i, it would be possible to ensure the minimum criterion of quality :

$$Q^* = \arg\inf_{Q} F(H, Q), \tag{2}$$

where F - is some operator. It is quite clear that the condition (2) can be complied with in the case we can correctly select the most efficient countermeasure Z_i for the detected jamming signal H_i . Once we know the types of interference which can affect the radar and actions to reduce efficiency of such impact, we would be in position to create decision procedures which allow each class of jamming signals H_i to find correspondence to some set of $X = \{x\}$ features which characterize it and compensate Z_i impact. Thus:

$$\begin{cases} H_i = F_i(x_1, x_2, \ldots, x_n) \\ Z_i = L(H_i), \end{cases} \tag{3}$$

where L - is come linear operator. Here, H_i jamming signals are determined a priori on the set of $X = \{x\}$ features. Let us assume that that the code of $H = \{H_i\}$ jamming signals and dictionary of $X = \{x\}$ features are optimal as implied in (2), i.e. the way to divide the feature space is selected so that the optimal value of quality criterion is achievable. The task then would be to find out such a rule that would provide for maximum efficiency in recognizing a jamming signal at specified criteria and restrictions [2]. In the general case, $X = \{x\}$ features can have deterministic, probabilistic, logical and structural character [3]. Relations (3) will take this or that shape depending on the type of features used. As far as determined features go, classes can be specified either by values of object features assigned a priory to each class or by coordinate values of some points in the feature space which are accepted as class templates. It is quite understandable that probabilistic features would have class descriptions in the form of probability-density functions $f_i(x_1, x_2, \ldots, x_n)$ of feature values $x_1, x_2, \ldots, x_n$, provided the object belongs to class H_i. When logical features of $x_i = \{1, 0\}$, $\forall i = \overline{1, q}$ type are used, then the class descriptions will be represented by Boolean relations between features and classes. Therefore it is possible to construct some or other decision rules which would enable use to

978-1-4244-3861-7/09/$25.00 ©2009 IEEE

recognize the jamming in progress from denumerable qnty of possible ones depending on types of features which compose the feature space $X = \{x\}$. Selecting the number and types of features is the objective of developing the feature space which can be solved based on study of radar operation under conditions of specified set of jamming features $H = \{H\}$. Apart from this, the synthesis of decision rules for recognition is possible only after the issue of feature space discretization is resolved to divide it into classes. Following specifications to develop the radar, the code of jamming classes $\{H\}$ will be reasonably regarded as denumerable set of output $q : H_i \in \{H\}, \forall i = \overline{1,q}$. It is clear that the feature space $X \subset R^n$ will be developed based on actual capacities of a definite radar to obtain information. Resolving the issue of discretization here is understood as such way to divide the feature space X that represents it as the sum of its pairwise disjoint subspaces $\bigcap_{i=1}^{q} X_i = \emptyset$, where $\emptyset$ - empty set, while closures $[X_1], [X_2], \ldots, [X_q]$ do not contain isolated points. In this regard, it is an essential requirement that features be consistent and non-contradictory whereby the convex feature space is developed and its structure closed at R^n. Thus, at $n = q$ each i-n class of H_i jamming can be related to some i-n detector X_i with output signal $x_i = \{1, 0\}$ $\forall i = \overline{1,q}$. Let us assume that at some j-n moment the radar is affected by H_i jamming signal, while there is a vector of observations $X_j = (x_{j1}, x_{j2}, \ldots, x_{jn})^T \in R^n$ developed at the exit of interference locator bar.

Recognition of H_i jamming is supposed to be implemented by the following decision rule :

$$\forall j, H_i : \sum_{k=1}^{n} (x_{jk} - h_{ik}) = 0, \tag{4}$$

where$H_i = (h_{i1}, h_{i2}, \ldots, h_{in})$ - vector of jamming image specified a priori.

Applying this approach to the airborne monopulse direction finder would significantly enhance its potential in conditions of jamming in progress.

II. CROSS-POLARIZATION JAMMING

Fig.1. presents the time variable picture for the plane of linear polarized plane interference signal in the base of airborne direction finder receiving antenna.

Polarization jamming acted on the monopulse direction finder with reflector receiving linear polarization antenna. In Fig.2., curves 2,3 & 4 represent the error signal of signal source autotracking. Curve 2 demonstrates loss in automatic tracking due to impact of polarization interference. After collation curve 2 Fig.2 and line 1 Fig.1. we can see loss in automatic tracking occurs in some ϵ-vicinities around point $90°$. Curve 3 is an autotracking error signal with no interference. Curve 4 characterizes locator operation after the jamming is recognized and measures to protect from its impact are taken. As can be seen from Fig.2, the airborne direction finder consistently tracks the designated target. Necessary to notice: in the experimental research of the

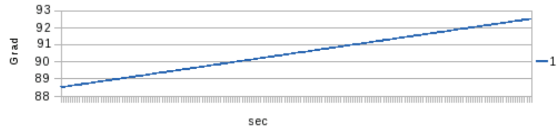

Fig. 1. Inclination angle of jamming polarization plane

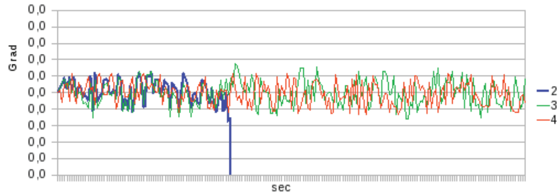

Fig. 2. The errors of angle tracking for cross-polarization jamming

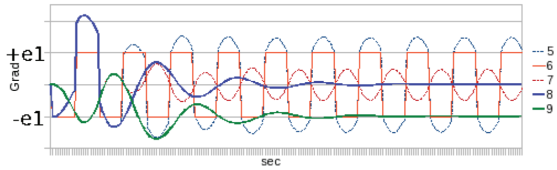

Fig. 3. The error signal and boresight position for blinking jamming

airborn locator on firing range was use real antiaircraft missile seeker. As a resalt - experimental confirm: deployment of new technical solutions developed by the author will make it possible to eliminate the danger from polarization interference impact and allows to refrain from deployment of costly phased array antenna in airborne radars of guided ground-to-air missiles.

III. BLINKING JAMMING

Operation of monopulse locator against blinking interference is shown in Fig.3. Jammers were placed $\pm 2°$ relative to the boresight of airborne locator receiving antenna. Interference control input is represented by curve 5, while boresight position by curve 7. The protected mode is determined by curves 8 and 9 which represent the error signal and boresight position respectively. Fig.3 (curve 9) shows that the airborne locator successfully recognized jamming and then implemented autonomous selection of target located at point $-e_1$ off-boresight angle of its receiving antenna.

IV. THE JAMMING IMMUNITY INDICATOR

To implement proper comparison of angle tracking accuracy for airborne locator under no-interference conditions against those with presence of active polarization and blinking jamming, it is necessary to provide for equal analysis intervals $[0, T]$ in each experiment. Compliance with this requirement and application of procedure (1) with subsequent logarithmation gives a visual presentation on jamming immunity of its operations in cases under consideration.

The jamming immunity indicator Q_j was calculated

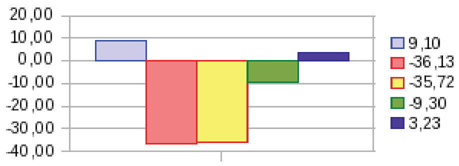

Fig. 4. Indicator of radar interference immunity

on the following formula:

$$Q_j = \ln\left(\frac{q_c}{q_j}\right) \tag{5}$$

After change in (1) determined on length $[0, T]$ integral to N final amounts, at condition of the constant step to sampling $\triangle t = const$, formula (1) will be copy in the manner of:

$$\hat{Q}_j = \sum_{i=1}^{N} e_{ji}^2 \triangle t, \qquad \forall j = 0, M \tag{6}$$

For behaviour of the benchmark analysis will be correct to use the relative indicator of jamming immunity. As master parameter for jamming immunity indicator will sensibly use the value of quality automatic tracing without jamming and interference signal in the base of airborne direction finder receiving antenna must be with main polarization. Puted master parameter in numerator, and parameter of the under investigation event in denominator, after reduction on $\triangle t$ we can write:

$$q_j = \frac{\sum\limits_{i=1}^{N} e_{0i}^2}{\sum\limits_{i=1}^{N} e_{ji}^2}, \qquad \forall j = 0, M \tag{7}$$

where $j = 0$ - zero-index for automatic tracking without jamming. However, calculation of the jamming immunity indicator by (7) will not give the beliefs about border of its applicability. Consequently, necessary to compare the results of the formula (7) with certain border value C. Taking into consideration that fact of target destruction by splinter combat charge with probability $P \geq A = const$ will possible if missile miss h will be not more then radius combat splinter R, which does this probability of the destruction possible, we can write:

$$h \leq R \tag{8}$$

Thereby if it is determined a priori missile velocity V_m and missile time constant t_s as a control system, then for greatly possible error of angle tracking in degree we can possible to write:

$$e_c = \frac{180}{\pi} \arctan\left[\frac{R}{V_m t_s}\right] \tag{9}$$

After substitution (9) in (7) and after reduction $\triangle t$ in numerator and denominator, we can write:

$$q_c = \frac{\sum\limits_{i=1}^{N} e_{0i}^2}{\sum\limits_{i=1}^{N} e_c^2} \tag{10}$$

The result from differences formula (7) and formula (8) and is subsequent logarithmation gives formula for antijamming indicator for j-variant of the analysis to writeas:

$$Q_j = \ln \sum_{i=1}^{N} e_{ji}^2 - \ln \sum_{i=1}^{N} e_c^2 \tag{11}$$

The border value C for this will be of the form of:

$$C = N \sum_{i=1}^{N} e_c^2 \tag{12}$$

It can be seen from Fig.4. for example unprotected radar has positive values: $Q = 9,10$ indicator for polarization jamming and $Q = 3,23$ for blinking jamming. A radar with high jam-resistance is characterized by negative values. Besides, Fig.4. shows that application of authors new method to effect the angle tracking of polarization jamming in the airborne locator provides the value of jamming immunity indicator almost identical to this indicator value for autotracking under no-interference conditions $(-36,13 \approx -35,72)$.

V. CONCLUSIONS

Providing for jamming immunity is particularly vital for airborne locators with the active homing channel. First and foremost this applies to firing against low-altitude radar targets beyond the radar horizon. Practical implementation of described and other approaches and principles developed by the author will allow to achieve performance level well above that of the new air defence systems and ensure consistent selection of targets in group, non-sensitivity of airborne to polarization and repeat jamming, non-sensitivity to clutters reflected from ground (water) surface and to antipode-type interference.

REFERENCES

[1]. A.I. Leonov, K.I. Fomichev. Monopulse Radar. Artech House. 1986.

[2]. A.L. Gorelik. General Objective of Object & Phenomenon Recognition. Kybernetika, 1980.- 6, p.72-75.

[3]. A.L. Gorelik, V.A. Skripkin. Recognition methods. M: Vysshaya Shkola, 1989.